

shp_pypsa_workflow

April 30, 2025

```
[1]: import pypsa
import geopandas as gpd
from shapely.geometry import LineString
import pandas as pd

# Step 1: Load shapefiles
gdf_buses = gpd.read_file("qgis/buses_aut.shp").to_crs(epsg=4326) #EPSG:4326
↳ WGS 84 -- WGS84 - World Geodetic System 1984, used in GPS
gdf_lines = gpd.read_file("qgis/lines_aut.shp").to_crs(epsg=4326)

# Step 2: Initialize PyPSA network
n = pypsa.Network()
#n.set_snapshots(pd.date_range("2025-01-01", periods=1))

# Step 3: Add buses
gdf_buses = gdf_buses.dropna(subset=["geometry"])
gdf_buses["bus"] = [f"bus_{i}" for i in range(len(gdf_buses))]

for _, row in gdf_buses.iterrows():
    n.add("Bus", name=row["bus"], x=row.geometry.x, y=row.geometry.y)

# Build spatial index for bus lookup
bus_coords = {
    (round(row.geometry.x, 6), round(row.geometry.y, 6)): row["bus"]
    for _, row in gdf_buses.iterrows()
}

# Step 4: Add lines
for idx, row in gdf_lines.iterrows():
    line: LineString = row.geometry
    if not isinstance(line, LineString) or line.is_empty:
        continue
    coords = list(line.coords)
    if len(coords) < 2:
        continue

# Round coordinates to match bus key precision
```

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start = tuple(map(lambda v: round(v, 6), coords[0]))
end = tuple(map(lambda v: round(v, 6), coords[-1]))

bus0 = bus_coords.get(start)
bus1 = bus_coords.get(end)

if not bus0 or not bus1 or bus0 == bus1:
    continue

n.add(
    "Line",
    name=f"line_{idx}",
    bus0=bus0,
    bus1=bus1,
    length=row.geometry.length, # in degrees, convert to km if desired
    s_nom=100 # dummy capacity
)

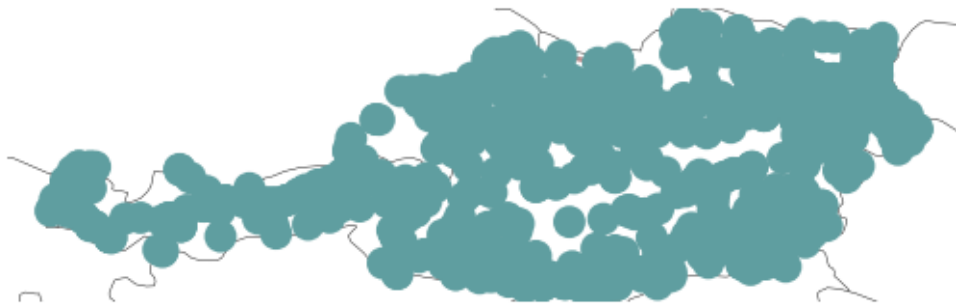
```

[2]: n.plot()

/home/bjoern/miniforge3/envs/pypsa-eur/lib/python3.12/site-packages/cartopy/mpl/feature_artist.py:144: UserWarning: facecolor will have no effect as it has been defined as "never".

warnings.warn('facecolor will have no effect as it has been ')

[2]: (<matplotlib.collections.PatchCollection at 0x7515e1808980>,
<matplotlib.collections.LineCollection at 0x7515e14d69f0>)



```

[3]: import pypsa
import matplotlib.pyplot as plt
import cartopy.crs as ccrs

# Plot the network
# Create a figure with GeoAxes using PlateCarree projection (standard lat/lon)
fig, ax = plt.subplots(

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figsize=(10, 10),
subplot_kw={"projection": ccrs.PlateCarree()}
)

n.plot(
    ax=ax,
    line_widths=2,
    bus_sizes=.0001,
    line_colors="blue",
    bus_colors="red",
    title="Austrian Electricity Grid",
)

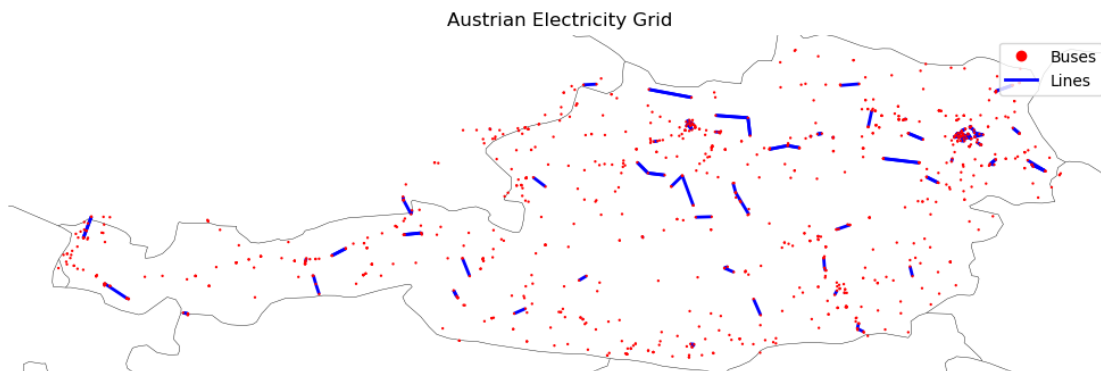
# Add legend manually
legend_elements = [
    plt.Line2D([0], [0], color='red', marker='o', linestyle='None',
    label='Buses'),
    plt.Line2D([0], [0], color='blue', lw=2, label='Lines'),
]
ax.legend(handles=legend_elements, loc='upper right')

plt.tight_layout()
plt.show()

```

/home/bjoern/miniforge3/envs/pypsa-eur/lib/python3.12/site-packages/cartopy/mpl/feature_artist.py:144: UserWarning: facecolor will have no effect as it has been defined as "never".

warnings.warn('facecolor will have no effect as it has been ')



```
[4]: len(n.lines)
```

```
[4]: 212
```

```
[5]: n.lines
```

```
[5]:
```

	bus0	bus1	type	x	r	g	b	s_nom	s_nom_mod	\
Line										
line_17	bus_773	bus_262		0.0	0.0	0.0	0.0	100.0	0.0	
line_20	bus_877	bus_773		0.0	0.0	0.0	0.0	100.0	0.0	
line_23	bus_831	bus_877		0.0	0.0	0.0	0.0	100.0	0.0	
line_31	bus_168	bus_131		0.0	0.0	0.0	0.0	100.0	0.0	
line_40	bus_777	bus_543		0.0	0.0	0.0	0.0	100.0	0.0	
...	...	...	...	...	...	...	...	...	...	...
line_1699	bus_913	bus_816		0.0	0.0	0.0	0.0	100.0	0.0	
line_1700	bus_913	bus_816		0.0	0.0	0.0	0.0	100.0	0.0	
line_1707	bus_789	bus_826		0.0	0.0	0.0	0.0	100.0	0.0	
line_2034	bus_903	bus_240		0.0	0.0	0.0	0.0	100.0	0.0	
line_2035	bus_903	bus_240		0.0	0.0	0.0	0.0	100.0	0.0	

	s_nom_extendable	...	v_ang_min	v_ang_max	sub_network	x_pu	\
Line		...					
line_17	False	...	-inf	inf		0.0	
line_20	False	...	-inf	inf		0.0	
line_23	False	...	-inf	inf		0.0	
line_31	False	...	-inf	inf		0.0	
line_40	False	...	-inf	inf		0.0	
...	...	...	...	...	...	...	...
line_1699	False	...	-inf	inf		0.0	
line_1700	False	...	-inf	inf		0.0	
line_1707	False	...	-inf	inf		0.0	
line_2034	False	...	-inf	inf		0.0	
line_2035	False	...	-inf	inf		0.0	

	r_pu	g_pu	b_pu	x_pu_eff	r_pu_eff	s_nom_opt
Line						
line_17	0.0	0.0	0.0	0.0	0.0	0.0
line_20	0.0	0.0	0.0	0.0	0.0	0.0
line_23	0.0	0.0	0.0	0.0	0.0	0.0
line_31	0.0	0.0	0.0	0.0	0.0	0.0
line_40	0.0	0.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...
line_1699	0.0	0.0	0.0	0.0	0.0	0.0
line_1700	0.0	0.0	0.0	0.0	0.0	0.0
line_1707	0.0	0.0	0.0	0.0	0.0	0.0
line_2034	0.0	0.0	0.0	0.0	0.0	0.0
line_2035	0.0	0.0	0.0	0.0	0.0	0.0

[212 rows x 31 columns]

```
[ ]: import geopandas as gpd
import pypsa
from shapely.geometry import LineString
```

```

# Load shapefiles
gdf_buses = gpd.read_file("qgis/buses_aut.shp").to_crs(epsg=4326)
gdf_lines = gpd.read_file("qgis/lines_aut.shp").to_crs(epsg=4326)

# Initialize PyPSA network
n = pypsa.Network()
n.set_snapshots(range(1)) # Minimal snapshot

gdf_buses = gdf_buses.dropna(subset=["geometry"])

# Add buses
for _, row in gdf_buses.iterrows():
    bus_id = str(row["id"])
    n.add("Bus",
          name=bus_id,
          x=row.geometry.x,
          y=row.geometry.y,
          v_nom=row.get("v_ref", 380),
          type=row.get("type", "substation"))

# Add lines
for _, row in gdf_lines.iterrows():
    line_geom: LineString = row.geometry
    if line_geom is None or line_geom.is_empty or len(line_geom.coords) < 2:
        continue

    bus0 = str(row["from_node"])
    bus1 = str(row["to_node"])
    if bus0 not in n.buses.index or bus1 not in n.buses.index:
        continue

    n.add("Line",
          name=str(row["id"]),
          bus0=bus0,
          bus1=bus1,
          length=row.get("length", line_geom.length) / 1000, # meters → km
          r=row.get("r", 0.05),
          x=row.get("x", 0.2),
          c=row.get("c", 0),
          s_nom=row.get("s_max", 100),
          type=row.get("type", None))

```

WARNING:pypsa.io:The following lines are already defined and will be skipped (use overwrite=True to overwrite): 11025.0

WARNING:pypsa.io:The following lines are already defined and will be skipped (use overwrite=True to overwrite): 11008.0

WARNING:pypsa.io:The following lines are already defined and will be skipped

```
(use overwrite=True to overwrite): 11026.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11008.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11031.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11014.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11025.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11021.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11022.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11024.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11030.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11026.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11029.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11032.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11033.0
WARNING:pypsa.io:The following lines are already defined and will be skipped
(use overwrite=True to overwrite): 11034.0
```

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[ ]: import pypsa
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import cartopy.crs as ccrs

# Plot the network
# Create a figure with GeoAxes using PlateCarree projection (standard lat/lon)
fig, ax = plt.subplots(
    figsize=(10, 10),
    subplot_kw={"projection": ccrs.PlateCarree()}
)

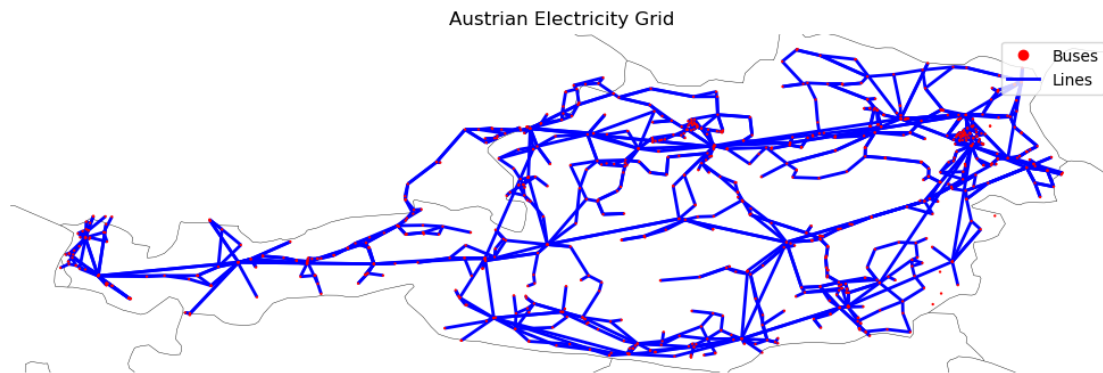
n.plot(
    ax=ax,
    line_widths=2,
    bus_sizes=.0001,
    line_colors="blue",
    bus_colors="red",
    title="Austrian Electricity Grid",
)
```

```
# Add legend manually
legend_elements = [
    plt.Line2D([0], [0], color='red', marker='o', linestyle='None',
        label='Buses'),
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ax.legend(handles=legend_elements, loc='upper right')

plt.tight_layout()
plt.show()
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/home/bjoern/miniforge3/envs/pypsa-eur/lib/python3.12/site-packages/cartopy/mpl/feature_artist.py:144: UserWarning: facecolor will have no effect as it has been defined as "never".

warnings.warn('facecolor will have no effect as it has been ')



```
[ ]: n.lines
```

```
[ ]:
```

	bus0	bus1	type	x	r	g	b	s_nom \
Line								
14001.0	14001.0	14003.0		0.3824	0.0557	0.0	0.0	205.196059
14002.0	14001.0	14003.0		0.3824	0.0557	0.0	0.0	205.196059
14003.0	14003.0	14001.0		0.4035	0.1690	0.0	0.0	89.547027
14004.0	14001.0	14002.0		0.4035	0.1690	0.0	0.0	89.547027
11030.0	11002.0	11005.0		0.4329	0.2024	0.0	0.0	89.500000
...	...	...	...	...	...	...	...	...
28065.0	28002.0	28015.0		0.3106	0.0256	0.0	0.0	777.344402
28068.0	28015.0	28050.0		0.3195	0.0210	0.0	0.0	914.522826
39121.0	39003.0	31027.0		0.2575	0.0151	0.0	0.0	1579.630337
31041.0	31027.0	36125.0		0.2574	0.0153	0.0	0.0	1372.962034
31042.0	31027.0	31020.0		0.0000	0.0000	0.0	0.0	0.000000

```

s_nom_mod  s_nom_extendable  ...  v_ang_max  sub_network  x_pu  r_pu  \

```

Line				...				
14001.0	0.0		False	...	inf		0.0	0.0
14002.0	0.0		False	...	inf		0.0	0.0
14003.0	0.0		False	...	inf		0.0	0.0
14004.0	0.0		False	...	inf		0.0	0.0
11030.0	0.0		False	...	inf		0.0	0.0
...	...		...	...	...	...	...	...
28065.0	0.0		False	...	inf		0.0	0.0
28068.0	0.0		False	...	inf		0.0	0.0
39121.0	0.0		False	...	inf		0.0	0.0
31041.0	0.0		False	...	inf		0.0	0.0
31042.0	0.0		False	...	inf		0.0	0.0

	g_pu	b_pu	x_pu_eff	r_pu_eff	s_nom_opt	c
Line						
14001.0	0.0	0.0	0.0	0.0	0.0	0.0000
14002.0	0.0	0.0	0.0	0.0	0.0	0.0000
14003.0	0.0	0.0	0.0	0.0	0.0	0.0000
14004.0	0.0	0.0	0.0	0.0	0.0	0.0000
11030.0	0.0	0.0	0.0	0.0	0.0	6.0483
...	...	...	...	...	...	...
28065.0	0.0	0.0	0.0	0.0	0.0	0.0000
28068.0	0.0	0.0	0.0	0.0	0.0	0.0000
39121.0	0.0	0.0	0.0	0.0	0.0	0.0000
31041.0	0.0	0.0	0.0	0.0	0.0	0.0000
31042.0	0.0	0.0	0.0	0.0	0.0	0.0000

[1394 rows x 32 columns]